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ELECTRONIC APPARATUS INCLUDING DISPLAY PROTECTIVE MEMBER

Abstract

An electronic device includes a flexible display panel and a flexible window disposed in a first direction, in which the flexible display panel displays an image, to protect the flexible display panel, and configured to be bent and unbent about a bending axis. The flexible window may include a first part which is located in an area of the flexible window relatively close to the bending axis, includes a first recess formed by the area of the flexible window being recessed, and is bent and unbent about the bending axis, and a second part which is thicker than the first part and is located in an area of the flexible window relatively distant from the bending axis, wherein the first part includes a reinforcement part formed to be oriented in a direction parallel to the bending axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application of International Application No. PCT/KR2023/017316, filed on Nov. 1, 2023, which is based on and claims priority to Korean Patent Application No. 10-2022-0143963, filed on Nov. 1, 2022, in the Korean Intellectual Property Office, and Korean Patent Application No. 10-2022-0184290, filed on Dec. 26, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND
1. Field
[0002] Various embodiments disclosed herein relate to an electronic device and, more particularly, to an electronic device including a display protection member.
2. Description of Related Art
[0003] Electronic devices require a small profile for portability and a large display area to provide a lot of information to a user. In order to reconcile the small profile and large display area of electronic devices, various form factors such as rollable, slidable, or foldable have emerged, moving away from the traditional rectangular bar-shaped form factor. Electronic devices with a form factor such as foldable require a flexible display that can be bent.
[0004] A display of an electronic device may include a display protection member, such as a window, that protects a panel layer by preventing foreign matter from entering from the top of the panel layer. A flexible display used in electronic devices with form factors such as rollable, slidable, or foldable requires a flexible window that can be bent or folded with the display panel.
[0005] A polymer material such as transparent polyimide may be used as the material of the flexible window, but the polymer material has low scratch resistance and resistance to repeated bending, so a glass material such as ultra-thin glass, which has both high hardness and flexibility, may be used as a window material.
[0006] Due to its thinness, ultra-thin glass is vulnerable to external impact, which may lead to breakage. When the thickness of a flexible is increased to improve the impact resistance of the window and prevent breakage, the flexibility and pliability of the flexible display window may be reduced. To solve this problem, the thickness of a first part, which is the part where bending occurs in a flexible display window, may be reduced. However, when different parts of glass have different thicknesses, warpage may occur during glass tempering (e.g., thermal and/or chemical tempering) due to an imbalance of stresses caused by the different thicknesses.
SUMMARY
[0007] Various embodiments of the disclosure may provide a display protection member with reduced warpage at a first part, and an electronic device including the same.
[0008] An electronic device according to various embodiments of the disclosure is an electronic device including a flexible display panel, and may include a flexible window disposed in a first

direction, in which the flexible display panel displays an image, to protect the flexible display panel, and configured to be bent and unbent about a bending axis. The flexible window may include a first part which is located in an area of the flexible window relatively close to the bending axis, includes a first recess formed by the area of the flexible window being recessed, and is bent and unbent about the bending axis, and a second part which is thicker than the first part and is located in an area of the flexible window relatively distant from the bending axis, wherein the first part includes a reinforcement part formed to be oriented in a direction parallel to the bending axis.
[0009] In various embodiments, the reinforcement part may include a rib protruding from the first part.
[0010] In various embodiments, the rib may be located at a center of the first part with respect to the bending axis.
[0011] In various embodiments, the reinforcement part may include reinforcement grooves recessed from a surface of the first part.
[0012] In various embodiments, the reinforcement grooves may be located to be symmetrically spaced apart from the center of the first part with respect to the bending axis.
[0013] In various embodiments, the electronic device may include a second recess formed on a surface opposite to the first recess with respect to a thickness direction of the flexible window.
[0014] In various embodiments, a boundary between the first part and the second part may have a tapered shape in a thickness direction.
[0015] In various embodiments, the flexible window may include a transparent polymer layer located in the recessed area of the first part.
[0016] In various embodiments, a thickness of the second part may exceed 30 micrometers.
[0017] In various embodiments, the flexible window may be manufactured by a method which includes a reinforcement part processing operation of processing and forming the reinforcement part by recessing a surface of the flexible window, a masking operation of locating a mask formed to cover a surface of the second part and expose an area corresponding to the first recess with respect to a surface of the glass layer, and an etching operation of etching the first recess.
[0018] A flexible window according to various embodiments of the disclosure is a flexible window disposed in a first direction, in which a flexible display panel displays an image in an electronic device including the flexible display panel, to protect the flexible display panel, and configured to be bent and unbent about a bending axis. The flexible window may include a first part which is located in an area of the flexible window relatively close to the bending axis, includes a first recess formed by the area of the flexible window being recessed, and is bent and unbent about the bending axis, and a second part which is thicker than the first part and is located in an area of the flexible window relatively distant from the bending axis, wherein the first part includes a reinforcement part formed to be oriented in a direction parallel to the bending axis.
[0019] In various embodiments, the reinforcement part may include a rib protruding from the first part.
[0020] In various embodiments, the rib may be located at a center of the first part with respect to the bending axis.
[0021] In various embodiments, the reinforcement part may include reinforcement grooves recessed from a surface of the first part.
[0022] In various embodiments, the reinforcement grooves may be located to be symmetrically spaced apart from the center of the first part with respect to the bending axis.
[0023] In various embodiments, the flexible window may include a second recess formed on a surface opposite to the first recess with respect to a thickness direction of the flexible window.
[0024] In various embodiments, a boundary between the first part and the second part may have a tapered shape in a thickness direction.
[0025] In various embodiments, the flexible window may include a transparent polymer layer located in the recessed area of the first part.

[0026] In various embodiments, a thickness of the second part may exceed 30 micrometers.
[0027] In various embodiments, the flexible window may be manufactured by a method which includes a reinforcement part processing operation of processing and forming the reinforcement part by recessing a surface of the flexible window, a masking operation of locating a mask formed to cover a surface of the second part and expose an area corresponding to the first recess with respect to a surface of the glass layer, and an etching operation of etching the first recess.
[0028] In various embodiments, puncture resistance of the second part and the first part may be 3 kgf or more.
[0029] In various embodiments, in the first part, the glass layer may be configured not to be broken during a pen drop test from a height of at least 6 cm.
[0030] In various embodiments, in the first part, the glass layer may be configured not to be broken by a pen tip during a pen pressing test with a force of at least 0.3 kgf.
[0031] In various embodiments, in case that the flexible window is bent, a radius of curvature of the first part may be less than or equal to 0.5 mm.
[0032] According to various embodiments disclosed herein, the first part of the flexible window may include a rigid reinforcement part, thereby reducing warpage due to stress imbalance in the first part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0033] In describing the drawings, the same or similar reference signs may be used for the same or similar elements.
[0034] FIG. **1** is a block diagram of an electronic device in a network environment according to various embodiments.
[0035] FIGS. **2**A to **2**F illustrate a portable electronic device having an in-folding housing structure according to an embodiment.
[0036] FIG. **3**A is a plan view illustrating a flexible display of an electronic device according to various embodiments of the disclosure.
[0037] FIG. **3**B is a cross-sectional view illustrating a flexible display according to various embodiments.
[0038] FIG. **3**C is a cross-sectional view illustrating a flexible display according to various embodiments.
[0039] FIG. **4**A is a schematic view illustrating a stress applied during tempering of a flexible window according to a comparative example.
[0040] FIGS. **4**B and **4**C are schematic views illustrating stresses applied during tempering of a flexible window according to embodiments of the disclosure.
[0041] FIG. **5**A is a side view illustrating the bent state of a flexible window according to a comparative example.
[0042] FIGS. **5**B and **5**C are side views illustrating the bent state of a flexible window according to embodiments of the disclosure.
[0043] FIG. **6** is a schematic diagram illustrating a manufacturing process of a flexible window according to various embodiments.
[0044] FIGS. **7**A to **7**C are cross-sectional views illustrating a flexible window according to various embodiments.
[0045] FIG. **8** is a perspective view of a test device for first evaluation, second evaluation, and third evaluation and a view showing the test device according to an embodiment.
[0046] FIG. **9** is a graph showing a minimum surface compressive stress that a glass layer must have to substantially prevent breakage of a glass layer depending on the drop height of a pen input

device according to an embodiment.
[0047] FIG. **10** is a graph showing the drop height of a pen input device that causes breakage of a glass layer depending on a thickness of the glass layer as a first evaluation of a laminate including the glass layer in FIG. **3** according to an embodiment.
[0048] FIG. **11** is a view illustrating a test device and a glass layer in conjunction with a first evaluation, and a view illustrating a test device and a glass layer in conjunction with a second evaluation according to an embodiment.
[0049] FIG. **12** is a graph showing the drop height of a pen input device that causes breakage of a glass layer depending on the thickness of the glass layer as a result of a first evaluation using a test device, and a graph showing a pressing load through the pen input device that causes breakage of the glass layer depending on the thickness of the glass layer as a result of a second evaluation using the test device, according to an embodiment.
[0050] FIG. **13** is a graph showing a breaking load depending on the thickness of a glass layer as a result of a third evaluation using a test device (e.g., a puncture resistance evaluation based on the ASTM F1306-16 standard), according to an embodiment.
[0051] FIG. **14** is a cross-sectional view illustrating the glass layer in the folded state of the electronic device, and a graph showing the minimum radius of curvature that allows the glass layer to be bent without breaking depending on the thickness of the glass layer, according to an embodiment.
DETAILED DESCRIPTION
[0052] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input 1module **150**, a sound output 1module **155**, a display 1module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).
[0053] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified

function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0054] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display 1module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0055] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thererto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0056] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0057] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input 1module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0058] The sound output 1module **155** may output sound signals to the outside of the electronic device **101**. The sound output 1module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0059] The display 1module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display 1module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display 1module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0060] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input 1module **150**, or output the sound via the sound output 1module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0061] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0062] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0063] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0064] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0065] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0066] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0067] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0068] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0069] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.
[0070] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.
[0071] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.
[0072] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).
[0073] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service

related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0074] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0075] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0076] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0077] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium

and where the data is temporarily stored in the storage medium.
[0078] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
[0079] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.
[0080] According to various embodiments, a portable electronic device (e.g., the electronic device **101** in FIG. **1**) may have a foldable housing that is divided into two housings about a folding axis. A first part of a display (e.g., a flexible display) may be disposed in a first housing, and a second part of the display may be disposed in the second housing. The foldable housing may be implemented in an in-folding manner in which the first part and the second part face each other when the portable electronic device is folded. Alternatively, the foldable housing may be implemented in an out-folding manner in which the first part and the second part face each other in opposite directions when the portable electronic device is folded. A surface on which the first part and the second part of the display are disposed may be defined as the front surface of the portable electronic device, the opposite surface may be defined as the rear surface of the portable electronic device, and a surface surrounding the space between the front surface and the rear surface may be defined as the side surface of the portable electronic device.
[0081] According to the embodiments described herein, in-folding, in which the first part of the display in the first housing is disposed to face the second part of the display in the second housing when the display of the portable electronic device is folded, has been illustrated and described as an example. However, according to an embodiment, the same may be applied to out-folding in which the first part of the display in the first housing is disposed to face an opposite direction to the second part of the display in the second housing when the display is folded. In addition, the embodiments may be applied to a multi-foldable electronic device in which in-folding and in-folding are combined, in-folding and out-folding are combined, and out-folding and out-folding are combined.
[0082] FIGS. **2**A to **2**F illustrate a portable electronic device **200** having an in-folding housing structure according to an embodiment. Specifically, FIGS. **2**A and **2**B illustrate a front view of a foldable portable electronic device (hereinafter, simply an electronic device) in an unfolded, flat, or open state, FIG. **2**C illustrates a rear view of the electronic device in a folded or closed state, and FIG. **2**D illustrates a rear view of the electronic device in an unfolded state, FIG. **2**E illustrates the front surface in a partially folded state (alternatively, a partially unfolded state, or an intermediate state (a freestop state) between a fully folded state and a fully unfolded state, and FIG. **2**F is an

exploded perspective view of the electronic device.
[0083] Referring to FIGS. **2**A to **2**F, the portable electronic device **200** (e.g., the electronic device **101** in FIG. **1**) may include a first housing **210**, a second housing **220**, a hinge assembly **240** connecting the first housing **210** to the second housing **220** such that the second housing **220** is rotatable relative to the first housing **210**, a flexible or foldable display **299** disposed in a space formed by the foldable housings **210** and **220**, and a sensor module (e.g., the sensor module **176** in FIG. **1**).
[0084] The display **299** may be disposed from the first housing **210** to the second housing **220** across the hinge assembly **240**. The display **299** may be divided into a first display area **211**, disposed in the inner space of the first housing **210**, and a second display area **221**, disposed in the inner space of the second housing **220**, with respect to the folding axis A. The sensor module (e.g., an illuminance sensor) may be disposed below a sensor area (or light-transmitting area) **242***a* of the first display area **211** when viewed from the front. The location and/or size of the sensor area **242***a* in the first display area **211** may be determined by the location and/or size of the illuminance sensor disposed below the sensor area. For example, the size (e.g., diameter) of the sensor area **242***a* may be determined based on the field of view (FOV) of the illuminance sensor. In an embodiment, the sensor area **242***a* may be configured to have a lower pixel density and/or lower wiring density than surroundings thereof in order to improve light transmittance. In some embodiments, the display **299** may include a protective layer **298** that includes a transparent material and protects a panel layer from external foreign matter and external impacts.
[0085] The hinge assembly **240** may be implemented in an in-folding manner such that the two display areas **211** and **221** face each other when the portable electronic device **200** switches from an unfolded state (e.g., the state in FIG. **2**A) to a folded state (e.g., the state in FIG. **2**C). For example, when the electronic device **200** is in the unfolded state, the two display areas **211** and **221** may face substantially the same direction. As the electronic device **200** switches from the unfolded state to the folded state, the two display areas **211** and **221** may be rotated to face each other. The hinge assembly **240** may be configured such that the foldable housings **210** and **220** have resistance against rotation. When an external force that exceeds the resistance is applied to the foldable housings **210** and **220**, the foldable housing **210**, **220** may be rotated.
[0086] The state of the electronic device **200** may be defined based on an angle formed between the two display areas **211** and **221**. For example, the state of the electronic device **200** may be defined as an unfolded, flat, or open state when the angle between the two display areas **211** and **221** is about 180 degrees. When the angle between the two display regions **211** and **221** is between about 0 degrees and about 10 degrees, the state of the portable electronic device **200** may be defined as a folded or closed state. When the two display areas **211** and **221** form an angle greater than the angle in the folded state and smaller than the angle in the unfolded state (e.g., between about 10 degrees and 179 degrees), the state of the portable electronic device **200** may be defined as an intermediate state (in other words, a partially folded or partially unfolded state) as illustrated in FIG. **2**E.
[0087] An active area in which visual information (e.g., text, image, or icon) will be displayed on the display **299** may be determined based on the state of the portable electronic device **200**. For example, when the electronic device **200** is in an intermediate state, the active area may be determined to be the first display area **211** or the second display area **221**. The area with relatively less movement between the first display area **211** and the second display area **221** may be determined as the active area. For example, when a user grips one housing of the electronic device **200** with one hand and opens the other housing with a finger (e.g., thumb) of the same hand or with the other hand, the electronic device **200** may switch from a folded state to an intermediate state, and the electronic device **200** may thereby determine that the displayed area of the gripped housing (e.g., the housing with relatively less movement) is the active area. When the portable electronic device **200** is in the unfolded state, the entire area (e.g., both the first display area **211** and the second display area **221**) of the display **299** may be determined to be an active area.

[0088] According to various embodiments, the first housing **210** may include, in an unfolded state, a first surface (the first display area) **211** facing a first direction (e.g., a front direction) (the z-axis direction) and a second surface **212** facing a second direction (e.g., a rear direction) (the-z-axis direction) opposite to the first side **211**. The second housing **220** may include, in an unfolded state, a third surface (the second display area) **221** facing the first direction (e.g., the z-axis direction) and a fourth surface **222** facing the second direction (e.g., the −z-axis direction). The electronic device **200** may be operated in such a manner that, in the unfolded state, the first surface **211** of the first housing **210** and the third surface **221** of the second housing **220** face the same first direction (e.g., the z-axis direction) and that, in the folded state, the first surface **211** and the third surface **221** face each other. The electronic device **200** may be operated such that, in the unfolded state, the second surface **212** of the first housing **210** and the fourth surface **222** of the second housing **220** face the same second direction (e.g., the −z axis direction) and in the folded state, the second surface **212** and the fourth surface **222** face opposite directions.
[0089] According to various embodiments, the first housing **210** may include a first lateral frame **213**, which at least partially forms the exterior of the electronic device **200**, and a first rear cover **214**, which is coupled to the first lateral frame **213** and forms at least a part of the second surface **212** of the electronic device **200**. According to an embodiment, the first lateral frame **213** may include a first side surface **213***a*, a second side surface **213***b* extending from one end of the first side surface **213***a*, and a third side surface **213***c* extending from the other end of the first side surface **213***a*. According to an embodiment, the first lateral frame **213** may be formed in an oblong (e.g., square or rectangular) shape by the first side surface **213***a*, the second side surface **213***b*, and the third side surface **213***c*.
[0090] A part of the first lateral frame **213** may be formed of a conductor. For example, referring to FIG. **2**B, a part {circle around (f)} of the first side surface **213***a*, a part {circle around (d)} of the second side surface **213***b*, and a part {circle around (e)} of the third side surface **213***c* may be formed of a metal material. The conductors may be electrically connected to grip sensors (not shown) adjacent thereto and disposed in the inner space of the first housing **210**. A processor may measure capacitance formed between the conductor and a ground (e.g., a ground of a main printed circuit board) through the grip sensors, and may recognize, based on the measured capacitance value, that a dielectric (e.g., the finger, the palm, or the face) has been in proximity (or contact) with the first housing **210** and where the first housing **210** is in contact with the dielectric (e.g., the first side surface **213***a*, the second side surface **213***b*, or the third side surface **213***c*).
[0091] According to various embodiments, the second housing **220** may include a second lateral frame **223**, which at least partially forms the exterior of the electronic device **200**, and a second rear cover **224**, which is coupled to the second lateral frame **223** and forms at least a part of the fourth surface **222** of the electronic device **200**. According to an embodiment, the second lateral frame **223** may include a fourth side surface **223***a*, a fifth side surface **223***b* extending from one end of the fourth side surface **223***a*, and a sixth side surface **223***c* extending from the other end of the fourth side surface **223***a*. According to an embodiment, the second lateral frame **223** may be formed in an oblong shape by the fourth side surface **223***a*, the fifth side surface **223***b*, and the sixth side surface **223***c*.
[0092] A part of the second lateral frame **223** may be formed of a conductor. For example, referring to FIG. **2**B, a part {circle around (b)} of the fourth side surface **223***a*, a part {circle around (a)} of the fifth side surface **223***b*, and a part {circle around (c)} of the sixth side surface **223***c* may be formed of a metal material. The conductors may be electrically connected to a grip sensor (not shown) adjacent thereto and disposed in the inner space of the second housing **220**. The processor may measure capacitance formed between the conductor and the ground (e.g., the ground of the main printed circuit board) through the grip sensor and may recognize, based on the measured capacitance value, that a dielectric has been in proximity to (or contact with) the second housing **220** and where the second housing **220** is in contact with the dielectric (e.g., the fourth side surface

223*a*, the fifth side surface **223***b*, or the sixth side surface **223***c*).

[0093] According to various embodiments, the pair of housings **210** and **220** are not limited to the illustrated shapes and combinations, and may be implemented by combinations and/or coupling of other shapes or components. For example, the first lateral frame **213** may be integrally formed with the first rear cover **214**, and the second lateral frame **223** may be integrally formed with the second rear cover **224**.

[0094] According to various embodiments, the first rear cover **214** and the second rear cover **224** may be formed by at least one or a combination of at least two of, for example, coated or colored glass, ceramic, polymer, or metal (e.g., aluminum, stainless steel (STS), or magnesium)

[0095] According to various embodiments, the electronic device **200** may include a first protective cover **215** (e.g., a first protective frame or a first decorative member) coupled along an edge of the first housing **210**. The electronic device **200** may include a second protective cover **225** (e.g., a second protective frame or a second decorative member) coupled along an edge of the second housing **220**. According to an embodiment, the first protective cover **215** and the second protective cover **225** may be formed of a metal or a polymer material.

[0096] According to various embodiments, the electronic device **200** may include a sub-display **231** disposed separately from the display **299**. According to an embodiment, the sub-display **231** may be disposed to be at least partially exposed on the second surface **212** of the first housing **210** so as to display state information of the electronic device **200** when the electronic device **200** is folded. According to an embodiment, the sub-display **231** may be disposed to be visible from the outside through at least a partial area of the first rear cover **214**. In some embodiments, the sub-display **231** may be disposed on the fourth surface **222** of the second housing **220**. In this case, the sub-display **231** may be disposed to be visible from the outside through at least a partial area of the second rear cover **224**.

[0097] According to various embodiments, the electronic device **200** may include at least one among an input device **203**, sound output devices **201** and **202**, camera modules **205** and **208**, a key input device **206**, a connector port **207**, and a sensor module (not shown). In an embodiment, the sensor module (e.g., the sensor module **176** in FIG. **1**) and the camera **205** may be disposed below the display **299** when viewed from the front.

[0098] According to various embodiments, the electronic device **200** may be operated to maintain an intermediate state through the hinge assembly **240**. In this case, the electronic device **200** may control the display **299** such that different contents are displayed on a display area corresponding to the first surface **211** and a display area corresponding to the third surface **221**.

[0099] Referring to FIG. **2**F, the electronic device **200** according to various embodiments may include the first lateral frame **213**, the second lateral frame **223**, and the hinge assembly **240** rotatably connecting the first lateral frame **213** and the second lateral frame **223** to each other. According to an embodiment, the electronic device **200** may include a first support plate **2131** extending at least partially from the first lateral frame **213**, and a second support plate **2231** extending at least partially from the second lateral frame **223**. According to an embodiment, the first support plate **2131** may be integrally formed with the first lateral frame **213** or structurally coupled to the first lateral frame **213**. Similarly, the second support plate **2231** may be integrally formed with the second lateral frame **223** or structurally coupled to the second lateral frame **223**. According to an embodiment, the electronic device **200** may include the display **299** disposed to be supported by the first support plate **2131** and the second support plate **2231**. According to an embodiment, the electronic device **200** may include the first rear cover **214** coupled to the first lateral frame **213** and providing a first space between the first rear cover **214** and the first support plate **2131**, and the second rear cover **224** coupled to the second lateral frame **223** and providing a second space between the second rear cover **224** and the second support plate **2231**. In some embodiments, the first lateral frame **213** and the first rear cover **214** may be integrally formed. In some embodiments, the second lateral frame **223** and the second rear cover **224** may be integrally formed. According to an embodiment, the electronic device **200** may include the first housing **210** provided by the first lateral frame **213**, the first support plate **2131**, and the first rear cover **214**. According to an embodiment, the electronic device **200** may include the second housing **220** provided by the second lateral frame **223**, the second support plate **2231**, and the second rear cover **224**.

[0100] Although not shown, the hinge assembly **240** may include a first arm structure coupled to the first housing **210** (e.g., the first support plate **2131**), a second arm structure coupled to the second housing **220** (e.g., the second support plate **2231**), and a detent structure physically contacting the first arm structure and the second arm structure such that the first housing **210** and the second housing **220** have resistance to rotation. The foldable housings **210** and **220** may have resistance to rotation by a contact force of the detent structure (e.g., a force pushing the first arm structure and the second arm structure).

[0101] According to various embodiments, the electronic device **200** may include a first substrate assembly **261** (e.g., a main printed circuit board), a camera assembly **263**, a first battery **271**, or a first bracket **251**, which is disposed in the first space between the first lateral frame **213** and the first rear cover **214**. According to an embodiment, the camera assembly **263** may include multiple cameras (e.g., the camera modules **205** and **208** in FIGS. **2**A and **2**C), and may be electrically connected to the first substrate assembly **261**. According to an embodiment, the first bracket **251** may provide a support structure and improved rigidity for supporting the first substrate assembly **261** and/or the camera assembly **263**. According to an embodiment, the electronic device **200** may include a second substrate assembly **262** (e.g., a sub printed circuit board), an antenna **290** (e.g., a coil member), a second battery **272**, or a second bracket **252**, which is disposed in the second space between the second lateral frame **223** and the second rear cover **224**. According to an embodiment, the electronic device **200** may include a wiring member **280** (e.g., a flexible printed circuit board (FPCB)) which is disposed to extend from the first substrate assembly **261**, across the hinge assembly **240**, to multiple electronic components (e.g., the second substrate assembly **262**, the second battery **272**, or the antenna **290**) disposed between the second lateral frame **223** and the second rear cover **224**, and provides electrical connectivity.

[0102] According to various embodiments, the electronic device **200** may include a hinge cover **241** that supports the hinge assembly **240** and is disposed to be exposed to the outside when the electronic device **200** is folded, and to be invisible from the outside by being moved into the first and second spaces when the electronic device **200** is unfolded.

[0103] According to various embodiments, the electronic device **200** may include the first protective cover **215** coupled along an edge of the first lateral frame **213**. According to an embodiment, the electronic device **200** may include the second protective cover **225** coupled along an edge of the second lateral frame **223**. In the display **299**, the edge of the first display area **211** may be protected by the first protective cover **215**. The edge of the second display area **221** may be protected by the second protective cover **225**. Protective caps **235** may be disposed in an area corresponding to the hinge assembly **240** to protect bending parts in the edge of the display **299**.

[0104] FIG. **3**A is a plan view illustrating a flexible display **300** of an electronic device according to various embodiments of the disclosure.

[0105] FIG. **3**B is a cross-sectional view illustrating the flexible display **300** according to various embodiments.

[0106] FIG. **3**C is a cross-sectional view illustrating the flexible display **300** according to various embodiments.

[0107] FIGS. **3**B and **3**C are cross-sectional views taken in the X-X′ direction in FIG. **3**A.

[0108] Referring to FIG. **3**A, a flexible display **300** (e.g., the display **299** in FIGS. **2**A to **2**F) of an electronic device (e.g., the electronic device **101** in FIG. **1** or the foldable electronic device **200** in FIGS. **2**A to **2**F) may include a first part **304** and a second part **305** located in an area other than the first part **304**. The first part **304** may be a part where the flexible display **300** can be bent and

unbent. For example, the flexible window **302** may be bent and unbent about a bending axis **303**.
[0109] Referring to FIGS. **3**B and **3**C, the flexible display **300** may include a flexible display panel **301** and a flexible window **302**. The flexible window **302** may be a member disposed in an image display direction (e.g., the z-axis direction) of the flexible display panel **301** to protect the flexible display panel **301**. In various embodiments, the flexible window **302** may include a transparent layer (e.g. glass layer **310**) having a transparent material. The transparent material may include glass and/or transparent polymer. The glass layer **310** may include, for example, a tempered glass material that has been strengthened by a thermal or chemical method. The tempered glass material may have high hardness and stiffness to effectively protect the flexible display panel **301** from external damage, such as penetration and crushing. In various embodiments, the glass layer **310** may include a first part **304**, which can be bent and unbent, and a second part **305**, which cannot be bent. In various embodiments, the glass layer **310** of the second part **305** may have a thickness of 30 micrometers or more, and thus may have high resistance to external forces, such as writing by a stylus pen or dropping of the stylus pen thereon.
[0110] The first part **304** of the flexible window **302** may have a first recess **311** formed to assist bending of the first part **304**. The first recess **311** may be formed, for example, in the glass layer **310**. The first recess **311** may be formed by recessing a surface (e.g., a surface facing an opposite direction to the z-axis) of the glass layer **310** of the first part **304**, and may be a part having a thickness less than the thickness of the second part **305**. By having a smaller thickness than the second part **305**, the first recess **311** may have lower stiffness and higher flexibility around the bending axis **303** than the second part **305**, and thus may allow the first part **304** of the flexible window **302** to be easily bent and unbent about the bending axis **303** during bending and unbending of the flexible display **300**. In various embodiments, the first recess **311** may have an aspect ratio (e.g., a dimension in the y-direction and a dimension in the z-direction) of 5:1 or greater.
[0111] In various embodiments, the first recess **311** may have a tapered shape that forms an angle with respect to the thickness direction (the z-axis direction). The tapered shape may smooth out the change in thickness from the second part **305** to the first recess **311**. Thus, optical distortion or scattering that may be caused by the thickness difference between the second part **305** and the first recess **311** may be mitigated, thereby improving the image quality of the flexible display **300**. In various embodiments, corners of the first recess **311** may have a round shape having the radius of curvature, R. Since the corner has a round shape, stress concentration on the corners of the first recess **311** may be mitigated.
[0112] A reinforcement part **313** may be formed in the first part **304** of the flexible window **302**. The reinforcement part **313** may be a part that reinforces the first part **304** by extending from the first part **304** in the direction of the bending axis **303** (e.g., in the x direction). Referring to FIG. **3**B, in various embodiments, the reinforcement part **313** may include a rib **313***b* protruding from the first part **304**. Referring to FIG. 3C, in various embodiments, the reinforcement part **313** may include a reinforcement groove **313***a* recessed from a surface of the first part **304**. FIGS. **3**B and **3**C illustrate embodiments in which two reinforcement parts **313** are formed, but the disclosure is not limited thereto, and it will be obvious to those skilled in the art that one or more reinforcement part **313** may be formed to reinforce the first part **304** against various stresses applied to the flexible window **302**, as described below.
[0113] In various embodiments, the reinforcement part **313** may be disposed on the face of the glass layer **310** so as to be close to or distant from the center of the first part **304**. For example, referring to FIG. **3**B, the reinforcement groove **313***a* may be disposed at a location relatively distant from the center of the first part **304**. Referring to FIG. **3**C, the rib **313***c* may be disposed at a location relatively close to the center of the first part **304**. An effect resulting therefrom will be described later.
[0114] In various embodiments, the flexible window **302** may include a transparent polymer layer

320. The transparent polymer layer **320** may be a member that fills the space between corrugations formed by locating the first recess **311** and the reinforcement part **313** of the first part **304**. Transparent polymer may include, for example, an acrylic-based resin such as acrylic urethane or poly-methyl-methacrylate (PMMA) and/or a material such as siloxane. The refractive index of the transparent polymer layer **320** may be equal or similar to the refractive index of the glass layer **310**. For example, the refractive index of the transparent glass may differ by less than 0.2 refraction index unit (RIU) from the refractive index of the glass layer **310**. Thus, the flexible window **302** including the transparent polymer layer **320** may prevent and/or minimize degradation of image quality due to refraction or scattering of light caused by the corrugations formed by the first recess **311** and the reinforcement part **313** in the glass layer **310**.

[0115] In various embodiments, the flexible window **302** may include a film layer **330**. The film layer **330** may be a film made of a polymer material having high hardness and flexibility to protect the surface of the glass layer **310** from impact and scratches. The material of the film layer **330** may include, for example, colorless polyimide (CPI), PET, or transparent polyurethane. Although not shown, in various embodiments, the film layer **330** may be attached to the surface of the flexible window **302** by means of a member such as a pressure-sensitive adhesive (PSA).

[0116] FIG. **4**A is a schematic diagram illustrating the expansion of each part due to a stress applied during tempering of a flexible window **2** according to a comparative example.

[0117] FIGS. **4**B and **4**C are schematic views illustrating the expansion of each part of a stress applied during tempering of the flexible window **302** according to embodiments of the disclosure.

[0118] The flexible window **2** according to the comparative example in FIG. **4**A is a flexible window **2** in which a first recess **11** is disposed in a first part **4** for flexibility of the first part but a reinforcement part is not disposed in the first recess **11**.

[0119] Referring to FIG. **4**A, the flexible window **2** according to the comparative example may be subjected to disproportionate strains E**1** and E**2** due to volume changes caused by thermal or chemical tempering, due to a thickness difference between and a second part **5** and the first recess **11** formed for the flexibility of the first part **4**. For example, in a thicker flat section **5**, the effect of volume change due to ionic infiltration or thermal shrinkage in the chemical tempering process is relatively small in the total volume, and a low strain El may occur. The thin first recess **11** is relatively more affected by a volume change and may therefore have a large strain E**2**. Due to the difference in strain described above, warpage may occur in the first recess **11** constrained by the second part **5**. For example, warpage may occur in the first recess **11** along the x-axis direction.

[0120] Referring to FIB. **4**B and **4**C, the flexible window **302** according to embodiments of the disclosure may reduce warpage in the x-axis direction as the reinforcement part **313** located in the first recess **311** provides additional stiffness to the first recess **311**. Thus, the occurrence of warpage due to a thickness difference and a strain disproportion may be reduced.

[0121] In addition, referring to FIG. **4**B, since a part where a rib is disposed in the first recess **311** is relatively thick compared with the first recess **311**, the volume change in the part where the rib is disposed may be smaller than those of other parts of the first recess **311**, and thus the strain E**3** in that part may also be relatively small. Therefore, the rib may mitigate strain difference due to a difference in the volume change of the first recess **311**.

[0122] In addition, referring to FIG. **4**C, a part where the reinforcement groove **313***a* is disposed in the first recess **311** is the periphery of the first recess **311**, and may be disposed so that the distribution of strain increases stepwise from E**1** of the second part **305** to E**2** of the reinforcement groove **313***a* via the strain E**3** of the first recess **311**, thereby mitigating the occurrence of warpage due to a sudden change in strain such as the comparative example in FIG. **4***a*.

[0123] FIG. **5**A is a side view illustrating the bent state of the flexible window **2** according to a comparative example.

[0124] FIGS. **5**B and **5**C are side views illustrating the bent state of the flexible window **302** according to embodiments of the disclosure.

[0125] The flexible window **2** according to the comparative example in FIG. **5**A is a flexible window **2** in which the first recess **11** is disposed in the first part **4** for flexibility of the first part **4** but the reinforcement part **313** is not located in the first recess **11**.
[0126] Referring to FIG. **5**A, when the flexible window **2** according to the comparative example is bent along a bending axis, the bending moment of the first part **4** is concentrated in the center of the first recess **11**, and thus the radius of curvature R**1** of the central part of the first part **4** may be smaller than the radius of curvature R**2** of the peripheral part thereof. The smaller radius of curvature R**1** may cause excessive strain in the film layer **30** disposed on the surface of glass, thereby causing whitening of the polymer material of the film layer **30** by plastic deformation, or causing the film layer **30** to detach from the glass layer **10**.
[0127] Referring to FIG. **5**B, according to an embodiment of the disclosure, the reinforcement groove **313***a* may be spaced apart from the center of the first part **304**. A part of the first part **304** where the reinforcement groove **313***a* is formed may have relatively low stiffness with respect to a bending moment about the bending axis **303**. Therefore, the central part of the first part **304** may have relatively high stiffness. Since the stiffness of the central part is relatively high and the stiffness of the part where the reinforcement groove **313***a* is disposed is relatively low, the bending moment may be distributed from the central part of the first part **304** to the periphery where the reinforcement groove **313***a* is disposed, compared with the comparative example in FIG. **5**A. Therefore, the radius of curvature R**3** of the central part of the first part **304** may be increased compared with the radius of curvature R**1** in the comparative example, and whitening and detachment of the film layer **330** due to the concentration of strain on the central part may be prevented or reduced.
[0128] Referring to FIG. **5**C, according to an embodiment of the disclosure, the rib may be disposed in or adjacent to the central part of the first part **304**. The part where the rib is disposed may have relatively higher stiffness with respect to the bending moment about the bending axis **303** than other parts of the first part **304**. Therefore, the bending moment may be distributed from the central part of the first part **304** to the periphery thereof, and thus whitening or detachment of the film layer **330** due to the concentration of strain on the central part may be prevented and reduced as described above.
[0129] FIG. **6** is a schematic diagram illustrating a manufacturing process of the flexible window **302** according to various embodiments.
[0130] Referring to FIG. **6**, the flexible window **302** according to various embodiments reinforcement part may be manufactured by a reinforcement part processing operation S**1**, a masking operation S**2**, and an etching operation S**3**.
[0131] The reinforcement part processing operation S**1** may be an operation of forming the reinforcement part **313**, such as the reinforcement groove **313***a*, on the surface of the glass layer **310**. A mechanical processing means, such as a grinder, and/or a chemical processing means using a masking and etching solution may be used as a means for forming the reinforcement groove **313***a*.
[0132] The masking operation S**2** may be an operation of locating a mask, which exposes an area corresponding to the first recess **311**, on the surface of the glass layer **310** on which the reinforcement part **313** is processed. The mask may include, for example, a masking film, print, or photoresist.
[0133] The etching operation S**3** may be an operation of etching the area exposed by the mask to form the first recess **311**. The etching may be performed by immersing the glass layer **310** in an etching solution and/or by spraying an etching solution onto the surface of the glass layer **310**. The etching solution may include an acidic etching solution and/or a basic etching solution containing components such as hydrofluoric acid or fluoride.
[0134] When etching, the etching solution may etch not only the first recess **311** but also the reinforcement part **313**. Accordingly, in the reinforcement part processing operation S**1**, the

reinforcement part **313** may be processed in consideration of the etching in the etching operation S**3**. For example, in the reinforcement part processing operation S**1**, the reinforcement groove **313***a* may be formed to have a size smaller than the final size of the reinforcement groove **313***a*. Although not shown, it will be obvious to those skilled in the art that in the reinforcement part processing operation S**1**, a reinforcement rib **313***b* may be formed to have a size larger than the final size of the reinforcement rib **313***b*.

[0135] FIGS. **7**A to **7**C are cross-sectional views illustrating the flexible window **302** according to various embodiments.

[0136] Referring to FIG. **7**A, the reinforcement part **313** formed in the first part **304** of the flexible window **302** may have a tapered shape in the thickness direction (the z-axis direction). The taper may be a shape in which the width of the reinforcement part **313** gradually decreases in a direction away from the surface of the first recess **311**. Since the reinforcement part **313** has a tapered shape, the difference in thickness between the reinforcement part **313** and other parts of the first recess **311** becomes gradual, so degradation of image quality due to refraction or scattering occurring in the reinforcement part **313** may be prevented or reduced.

[0137] Referring to FIG. **7**B, the reinforcement part **313** formed in the first part **304** of the flexible window **302** may have a curved surface shape, for example, a semicircular shape. The curved surface shape may prevent stress from being concentrated on the corners of the reinforcement part **313**.

[0138] Referring to FIG. **7**C, the flexible window **302** may include a second recess **312** formed on a surface opposite to the first recess **311** with respect to the thickness direction (e.g., the z-axis direction) of the glass layer **310**. Since the second recess **312** is included, the thickness of the first part **304** may be easily reduced without making the aspect ratio of the first recess **311** excessively small (e.g., less than 5:1). Accordingly, any reduction in visibility or concentration of stress on the corners of the first recess **311** that may be caused by an excessively large aspect ratio of the first recess **311** may be reduced. The second recess **312** may include a reinforcement part **313** for further reinforcing the first part **304**. Although FIG. **7**A illustrates an embodiment in which the reinforcement part **313** is a rib **313***b*, it will be obvious to those skilled in the art that an embodiment in which the reinforcement groove **313***a* is formed in the second recess **312** is also possible. With regard to the reinforcement part **313** formed in the second recess **312**, reference may be made to the description of the reinforcement part **313** formed in the first recess **311** insofar as the description is not contradictory.

[0139] FIG. **8** is a perspective view **1301** of a test device **1300** for first evaluation, second evaluation, and third evaluation, and a view **1302** showing the test device **1300** according to an embodiment.

[0140] Referring to FIG. **8**, in an embodiment, the test device **1300** may include a clamping device (e.g., a manual clamping fixture) **1310** and a probe (e.g., a penetration probe) **1320**. A flexible material (e.g., a flexible film and/or a laminate) (e.g., the glass layer **310** in FIG. **3**, or a laminate including the glass layer **310**) may be disposed on the clamping device **1310**. The probe **1320** may be implemented to be linearly movable with respect to the clamping device **1310**. The probe **1320** may be moved linearly to apply a load to the flexible material disposed on the clamping device **1310**. When the probe **1320** breaks the flexible material, the load (e.g., a breaking load) may be detected by a detector (e.g., sensor) connected to the probe **1320**.

[0141] According to an embodiment, the clamping device **1310** may include a first plate **1311**, a second plate **1312**, and multiple supports **1313**.

[0142] According to an embodiment, the first plate **1311** may be a surface plate. A flexible material to be tested (e.g., the glass layer **310** in FIG. **3**, or a laminate including the glass layer **310**) may be disposed on the first plate **1311**. The first plate **1311** may include a support surface **1311***a* on which the flexible material to be tested is disposed. The support surface **1311***a* of the first plate **1311** may be substantially planar. The first plate **1311** may include an opening **1311***b* corresponding to the probe **1320**. The opening **1311***b* may reduce or prevent the effect of the first plate **1311** on detection of the strength of the flexible material. The opening **1311***b* may prevent the first plate **1311** from interfering with the linear movement of the probe **1320**.

[0143] According to an embodiment, the second plate **1312** may be spaced apart from the first plate **1311** in a direction in which the probe **1320** is moved linearly to press the flexible material (e.g., the glass layer **310** in FIG. **3** or a laminate including the glass layer **310**). The space between the first plate **1311** and the second plate **1312** may prevent the clamping device **1310** from interfering with the linear movement of the probe **1320**. The multiple supports **1313** may be disposed between the first plate **1311** and the second plate **1312**. The multiple supports **1313** may connect the first plate **1311** to the second plate **1312**. The second plate **1312** may be coupled to a body (not separately shown) included in the test device **1300**, and the first plate **1311** may be supported by the multiple supports **1313** and spaced apart from the second plate **1312** coupled to the body. The body of the test device **1300** is a substantial support body that stably supports elements such as the probe **1320** and the second plate **1312**, and may reduce vibration or shaking during operation of the test device **1300**.

[0144] According to an embodiment, the glass layer **310** in FIG. **3** or the laminate including the glass layer **310** may be disposed on the support surface **1311***a* of the first plate **1311**. According to an embodiment, the probe **1320** may include a pen input device (e.g., an electronic pen, a digital pen, or a stylus pen). The pen input device may be disposed on the test device **1300** while standing in a direction substantially perpendicular to the support surface **1311***a* of the first plate **1311**. When the test device **1300** is driven, a pen tip **1321** of the pen input device may press the glass layer **310** (see FIG. **3**) disposed on the first plate **1311**. It may be understood that moving the vertically upright pen input device in the test device **1300** to apply a load to the glass layer **310** (see FIG. **3**) is at least similar to a situation in which a user drops the pen input device, or a situation in which a user performs touch input using the pen input device.

[0145] According to an embodiment, the pen input device may substantially weigh about 5.6 grams.

[0146] According to an embodiment, the diameter D**1** of the pen tip **1321** included in the pen input device may be substantially about 0.3 mm.

[0147] According to an embodiment, the pen tip **1321** of the pen input device may be a ball tip made of tungsten carbide. The ball tip may be provided with a curved surface having a convex end.

[0148] According to various embodiments, the probe **1320** may be provided as an object that can substantially replace the pen input device.

[0149] According to an embodiment, the test device **1300** may follow the ASTM F1306-16 standard for evaluating puncture resistance of a flexible material in relation to the third evaluation. The probe **1320** may be provided in accordance with the ASTM F1306-16 standard. According to the ASTM F1306-16 standard, the support surface **1311***a* of the first plate **1311** may have a surface roughness of substantially about 200 μm. According to the ASTM F1306-16 standard, the diameter D**2** of the opening **1311***b* of the first plate **1311** may be substantially about 35 mm. In accordance with the ASTM F1306-16 standard, the distance H that the first plate **1311** and the second plate **1312** are spaced apart from each other in the linear direction of travel of the probe **1320** may be substantially about 76 mm.

[0150] According to an embodiment, the first evaluation and the second evaluation may be substantially performed by the test device **1300** according to the ASTM F1306-16 standard.

[0151] FIG. **9** is a graph showing a minimum surface compressive stress that the glass layer **310** (see FIG. **3**) must have to substantially prevent breakage of the glass layer **310** depending on the drop height of a pen input device according to an embodiment.

[0152] Referring to FIG. **9**, in an embodiment, the glass layer **310** may have a thickness of substantially about 0.14 mm or less than 0.14 mm. In order to reduce or prevent breakage of the glass layer **310** when a dropped pen input device impacts the glass layer **310**, the glass layer **310** may be implemented to have a compressive strength capable of withstanding a surface compressive stress of about 100 MPa or greater than 100 MPa.

[0153] FIG. **10** is a graph showing the drop height of a pen input device that causes breakage of the glass layer **310** depending on a thickness of the glass layer **310** as a first evaluation of a laminate including the glass layer **310** in FIG. **3** according to an embodiment.

[0154] According to an embodiment, the laminate including the glass layer **310** in FIG. **3** may include the glass layer **310**, a flexible display (e.g., the display **299** in FIG. **2**F), and an adhesive material (e.g., the impact absorption layer) between the glass layer **310** and the flexible display.

[0155] Referring to FIG. **10**, as the thickness of the glass layer **310** (see FIG. **3**) included in the laminate increase, the drop height of the pen input device that causes the glass layer **310** to break may also increase. In an embodiment, the thickness of the glass layer **310** may be limited to a value that allows an electronic device (e.g., the electronic device **200** in FIG. **2**A) to be bent with reduced bending stress without breakage when switching from an unfolded state to a folded state. In an embodiment, the thickness of the glass layer **310** may be limited to a value that reduces curvature and the radius of curvature as much as possible without breakage (or with reduced bending stress) when bent.

[0156] FIG. **11** is a view **1601** illustrating the test device **1300** and the glass layer **310** in conjunction with a first evaluation, and a view **1602** illustrating the test device **1300** and the glass layer **310** in conjunction with a second evaluation according to an embodiment.

[0157] Referring to FIG. **11**, in the first evaluation using the test device **1300**, a first protective film **1611** made of polymer may be disposed on (e.g., attached to) one surface **310***a* of the glass layer **310** by means of a first adhesive material (or first bonding material) **1612**. In the first evaluation using the test device **1300**, a second protective film **1621** made of polymer is may be disposed on (e.g., attached to) the other side **310***b* of the glass layer **310** by means of a second adhesive material (or a second bonding material) **1622**. The first protective film **1611** and the second protective film **1621** may be, for example, polyethylene terephthalate (PET) having a thickness of substantially about 50 μm. The first adhesive material **1612** and the second adhesive material **1622** may be, for example, a pressure sensitive adhesive (PSA) having a thickness of substantially about 25 μm or less than about 25 μm.

[0158] According to an embodiment, in the second evaluation using the test device **1300**, a protective film **1631** made of polymer may be disposed on (e.g., attached to) the bottom surface **310***b* of the glass layer **310** by means of an adhesive material (or a bonding material) **1632**. The protective film **1631** may be, for example, PET having a thickness of substantially about 200 μm. The adhesive material **1632** may be, for example, PSA having a thickness of about 10 μm, or less than 10 μm. In an embodiment, in the second evaluation using the test device **1300**, a protective film (e.g., a urethane jig) (not separately shown) of Hs30 hardness may be located to cover the bottom surface **310***b* of the glass layer **310**.

[0159] FIG. **12** is a graph **1701** showing the drop height of a pen input device (e.g., the probe **1320** in FIG. **8**) that causes breakage of the glass layer **310** (see FIG. **3**) depending on the thickness of the glass layer **310** as a result of the first evaluation using the test device **1300** (see FIG. **8**), and a graph **1702** showing a pressing load through the pen input device that causes breakage of the glass layer **310** depending on the thickness of the glass layer **310** as a result of the second evaluation using the test device, according to an embodiment. FIG. **13** is a graph showing a breaking load depending on the thickness of the glass layer **310** as a result of the third evaluation using the test device **1300** (see FIG. **8**) (e.g., a puncture resistance evaluation based on the ASTM F1306-16 standard), according to an embodiment.

[0160] According to an embodiment, in view of the result of the first evaluation disclosed in FIG. **12**, the glass layer **310** (see FIG. **3**) may be implemented such that the drop height of the pen input device which causes breakage of the glass layer **310** (see FIG. **3**) is about 6 cm or more than 6 cm.

[0161] According to an embodiment, in view of the result of the second evaluation disclosed in

FIG. **12**, the glass layer **310** may be implemented such that the pressing load through the pen input device which causes breakage of the glass layer **310** is about 0.3 kgf or more than 0.3 kgf.
[0162] According to an embodiment, in view of the result of the third evaluation disclosed in FIG. **13**, the glass layer **310** may be implemented such that the puncture resistance (e.g., breaking load or determination strength) is about 3 kgf, or greater than 3 kgf.
[0163] FIG. **14** is a cross-sectional view illustrating the glass layer **310** in the folded state of the electronic device **1** (see FIG. **1**), and a graph showing the minimum radius of curvature R that allows the glass layer **310** to be bent without breaking depending on the thickness T of the glass layer **310**, according to an embodiment.
[0164] Referring to FIG. **14**, bending stress generated in a bending area of the glass layer **310** may result from the collision of an increasing force (e.g., tensile stress) on one surface of the bending area and a decreasing force (e.g., compressive stress) on the other surface of the bending area. The bending stress generated in the bending area of the glass layer **310** may be proportional to the thickness T of the glass layer **310** and inversely proportional to the radius of curvature R. The radius of curvature R may be substantially based on the inner surface **310***a* of the glass layer **310** when the glass layer **310** is bent. In an embodiment, the thickness T of the glass layer **310** may be provided such that the glass layer **310** has a minimum radius of curvature R of about 5 mm, or greater than 0 mm and less than 5 mm that enables the glass layer **310** to be bent without breaking.
[0165] An electronic device according to various embodiments of the disclosure is an electronic device including a flexible display panel **301**, and may include a flexible window **302** disposed in a first direction, in which the flexible display panel **301** displays an image, to protect the flexible display panel **301**, and configured to be bent and unbent, wherein the flexible window **302** may include a first part **304** located in a first area of the flexible window **302** including a first recess **311** formed on a first surface of the flexible window **302** and is bent and unbent about the bending axis and a second part **305** which is thicker than the first part **304** and is located in a second area of the flexible window **302**, wherein the first part **304** may include a reinforcement part **313** formed on the surface of the first part **304** to be oriented in a direction vertical to the direction of the bending and unbending.
[0166] In various embodiments, the reinforcement part **313** may include a rib **313***b* protruding from the first part **304**.
[0167] In various embodiments, the rib **313***b* may be located at a center of the first part **304** with respect to the bending axis.
[0168] In various embodiments, wherein the reinforcement part **313** may include a reinforcement groove **313***a* recessed from a surface of the first part **304**.
[0169] In various embodiments, the reinforcement groove **313***a* may be located to be symmetrically spaced apart from the center of the first part **304** with respect to the bending axis.
[0170] In various embodiments, the electronic device may include a second recess **312** formed on a surface opposite to the first recess **311** with respect to a thickness direction of the flexible window **302**.
[0171] In various embodiments, a boundary between the first part **304** and the second part **305** has a tapered shape in a thickness direction.
[0172] In various embodiments, the flexible window **302** may include a transparent polymer layer **320** located in the recessed area of the first part **304**.
[0173] In various embodiments, a thickness of the second part **305** exceeds 30 micrometers.
[0174] In various embodiments, the flexible window **302** may be manufactured by a method comprising a reinforcement part **313** processing operation of processing and forming the reinforcement part **313** by recessing a surface of the glass layer, a masking operation of locating a mask formed to cover a surface of the second part **305** and expose an area corresponding to the first recess **311** with respect to a surface of the glass layer; and an etching operation of etching the first recess **311**.

[0175] A flexible window **302** disposed on a flexible display panel **301** of an electronic device, and configured to be bent and unbent about a bending axis, may comprise:
[0176] a first part **304** located in a first area of the flexible window **302** may include a first recess **311** formed on a first surface of the flexible window **302**, and is bent and unbent about the bending axis, and
[0177] a second part **305** which is thicker than the first part **304** and is located in a second area of the flexible window **302**,
[0178] and wherein the first part **304** may include a reinforcement part **313** formed on the surface of the first part **304** to be oriented in a direction vertical to the direction of the bending and unbending.
[0179] In various embodiments, the reinforcement part **313** may include a rib **313***b* protruding from the first part **304**.
[0180] In various embodiments, the rib **313***b* may be located at a center of the first part **304** with respect to the bending axis.
[0181] In various embodiments, wherein the reinforcement part **313** may include a reinforcement groove **313***a* recessed from a surface of the first part **304**.
[0182] In various embodiments, the reinforcement groove **313***a* may be located to be symmetrically spaced apart from the center of the first part **304** with respect to the bending axis.
[0183] In various embodiments, the electronic device may include a second recess **312** formed on a surface opposite to the first recess **311** with respect to a thickness direction of the flexible window **302**.
[0184] In various embodiments, a boundary between the first part **304** and the second part **305** has a tapered shape in a thickness direction.
[0185] In various embodiments, the flexible window **302** may include a transparent polymer layer **320** located in the recessed area of the first part **304**.
[0186] In various embodiments, a thickness of the second part **305** exceeds 30 micrometers.
[0187] In various embodiments, the flexible window **302** may be manufactured by a method comprising a reinforcement part **313** processing operation of processing and forming the reinforcement part **313** by recessing a surface of the glass layer, a masking operation of locating a mask formed to cover a surface of the second part **305** and expose an area corresponding to the first recess **311** with respect to a surface of the glass layer; and an etching operation of etching the first recess **311**.
[0188] In various embodiments, puncture resistance of the second part **305** and the first part **304** may be 3 kgf or more.
[0189] In various embodiments, the flexible window **302** may include a glass layer, and wherein in the first part **304**, the glass layer may be configured not to be broken during a pen drop test from a height of at least 6 cm.
[0190] In various embodiments, in the first part **304**, the glass layer may be configured not to be broken by a pen tip during a pen pressing test with a force of at least 0.3 kgf.
[0191] In various embodiments, in case that the flexible window **302** is bent, a radius of curvature of the first part **304** may be less than or equal to 0.5 mm.
[0192] In various embodiments, the flexible window **302** may include plurality of the second part **305** spaced apart by the first part **304**.
[0193] In various embodiments, at least some of reinforcement part **313** is formed on a surface of the second recess **312**.
[0194] In various embodiments, the reinforcement part **313** may have a rounded cross-sectional shape.
[0195] The embodiments disclosed herein, as disclosed in the specification and drawings, are merely provide particular examples in order to easily describe the technical matters according to the embodiments disclosed herein and help to understand the embodiments disclosed herein, and

are not intended to limit the scope of the embodiments disclosed herein. Therefore, it should be construed that the scope of the various embodiments disclosed herein includes, in addition to the embodiments disclosed herein, all modifications or variations derived from the technical ideas of the various embodiments disclosed herein.

Claims

1. A flexible window disposed on a flexible display panel of an electronic device, and configured to be bent and unbent about a bending axis, comprising: a first part located in a first area of the flexible window comprises a first recess formed on a first surface of the flexible window, and is bent and unbent about the bending axis; and a second part which is thicker than the first part and is located in a second area of the flexible window, and wherein the first part comprises a reinforcement part formed on the surface of the first part to be oriented in a direction vertical to the direction of the bending and unbending.
2. The flexible window of claim 1, wherein the reinforcement part comprises a rib protruding from the first part.
3. The flexible window of claim 2, wherein the rib is located at a center of the first part with respect to an axis of the bending and unbending of the first part.
4. The flexible window of claim 1, wherein the reinforcement part comprises a reinforcement groove recessed from a surface of the first part.
5. The flexible window of claim 4, wherein the reinforcement groove located to be symmetrically spaced apart from the center of the first part with respect to an axis of the bending and unbending of the first part.
6. The flexible window of claim 1, comprising a second recess formed on a surface opposite to the first recess with respect to a thickness direction of the flexible window, wherein at least some of reinforcement part is formed on a surface of the second recess.
7. The flexible window of claim 1, wherein a boundary between the first part and the second part has a tapered shape in a thickness direction.
8. The flexible window of claim 1, comprising a transparent polymer layer located in the recessed area of the first part.
9. The flexible window of claim 1, wherein a thickness of the second part exceeds 30 micrometers.
10. The flexible window of claim 1, wherein puncture resistance of the non-bending part and the bending part is 3 kgf or more.
11. The flexible window of claim 1, wherein in the bending part, a glass layer forming at least a part of the flexible window is configured not to be broken during a pen drop test from a height of at least 6 cm.
12. The flexible window of claim 1, wherein in the bending part, a glass layer forming at least a part of the flexible window is configured not to be broken by a pen tip during a pen pressing test with a force of at least 0.3 kgf.
13. The flexible window of claim 1, wherein in case that the flexible window is bent, a radius of curvature of the bending part is less than or equal to 0.5 mm.
14. An electronic device comprising: a flexible display panel configured to display an image in a first direction; and a flexible window according to claim 1, wherein the flexible window is disposed in the first direction of the flexible display.
15. A method for manufacturing a flexible window for protecting a flexible display panel, wherein the flexible window comprises a first part having a first recess and configured to be bent and unbent and a second part, the method comprising: processing and forming the reinforcement part by recessing a surface of the glass layer; locating a mask formed to cover a surface of the second part of the flexible window and expose an area corresponding to the first recess with respect to a

surface of the flexible window; and an etching operation of etching the surface of the flexible window to form the first recess.